

1. List down 5 types of problem commonly seen in nature which can be solved by supervised and unsupervised learning each. Mention which model will work best, what will be the benefits etc.

→ Supervised

-Deciding the outcome of a cricket match (random forest classifier as there is a lot of historical data and decision trees can handle huge data)

- predicting a batsman's score. (multiple linear regression) benefit: dream 11 :)

- a very common example I came across while learning: predicting real estate prices. Here multiple linear regression is best used.

- like we have speed cameras on highways, we can also install cameras to check if a driver is wearing a helmet. To use this we can use SVMs for binary classification.

- This idea I had got from LinkedIn once. We can predict if an email is spam or not. We can use logistic regression for this. Benefit is that we won't need to check unnecessary emails which consumes time. Note that features would be exclamation marks, words like click here, etc.

→ Unsupervised

- News detection. My mom watches shorts of a news channel on YouTube where the name of the video doesn't give any hint about the news and most of the time the news ends up being depressing. To avoid watching such unnecessary news, we can make clusters using the transcript of the videos to identify useful news. K-mean clustering is the way to go.

- making groups of cricket player by their batting style. This would help people who don't know much about cricket while making teams on Dream 11 :). Here also K-mean is the way to go.

- This one's already been done but- classifying songs on Spotify according to genre or moods like sad, happy, exciting, etc.

- this idea is from the time I was discussing with my friend once. At local mithai stores, we can see what each customer is most interested in buying, for example there'll be people who only purchase kaju katli from a sweet shop every 3 months approx. so for the kaju katli lovers there'll be an exclusive discount. To group customers according to their most bought sweet, we can give discounts.

- similar is a feature probably(not sure) used by OOT platforms- grouping viewers

according to the type of genre they follow the most and suggesting similar shows and movies.

2. Identify stock values where we can apply regression and apply linear regression to them. Is SLR fine to predict stock value if not what should be feature space for MLR.

→ - Stocks which are not event driven are needed for this. The stock should have good fundamentals like good management, satisfying ratios etc.

Stocks like Reliance, HDFC arent driven by news shocks.

Small cap stocks need to be avoided as they are very volatile and unpredictable.

I dont think SLR is the best fir because we cant predict based on only one feature obviously. For MLR features can be

- Prev closing price.
- Volumes
- 20 Day SMA
- RSI
- Prolly the standard deviation from the last 20 day or some window which would tell us the volatility.
- index rating
- P/E ratio (not sure)

One thing im not sure about is how can we still be sure about MLR as well? Because we are assuming a linear relationship here...